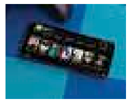
FB to open algorithms
FB willing to subject itself to more oversight to ensure its algorithms perform as intended. **https://dgit.in/nov21-39**



RTX ON with GeForce Now
The new RTX 3080 tier will allow 8 hour sessions and 120 FPS/1440p at 100$ per six months. **https://dgit.in/nov21-40**

MSI Spatium M480 2TB SSD

Great gen-on-gen upgrade

Price
₹33,998

80

FEATURES	80
PERFORMANCE	90
VALUE	75
DESIGN	75

MSI is a brand that comes to mind when you're talking about SSDs. As part of their initial launch into SSDs, the company has introduced several new PCIe Gen 4.0 and Gen 3.0 NVMe SSDs and the one we're reviewing is the MSI Spatium M480 2TB HS variant. But, does it have the chops to beat competition SSDs that have led the market for about a decade? That's what we're hoping to find out.

BUILD

MSI went with a Phison controller for the flagship SSD in their portfolio. In particular, it is using the Phison PS5018-E18 controller which would be the second gen E18 controller with several improvements that allow it to perform better in random data reads. The controller is powered by five ARM Cortex R5 cores of which three are the big cores and the remaining two are the little cores. It has eight NAND flash channels capable of running at 1600 MTps. The higher channel bandwidth is what helps with the random access performance.

The PS5018-E18 is paired with two SK Hynix 8Gb DDR4 DRAM chips, one of each side, for the cache and for the NAND it is using Micron's 96 layer 3D NAND TLC chips. Each of which is 256 Gigabytes.

FEATURES

One of the key features of the Phison E18 controller is that it offers up a pseudo-SLC cache of pSLC. To implement pSLC cache, the controller marks out a portion of the main NAND capacity as SLC cache. By utilising just one bit per cell in the NAND, it can write to the pSLC cache very quickly. With the Phison E18 controllers, you can take things further by implementing dynamic pSLC. This simply means, that instead of having just one portion of the NAND capacity being designated as the pSLC cache, the controller can mark up to 100% of the NAND capacity as flash.

Enough of the controller. Let's look at the heatsink. It is a two-part assembly with a massive chunk on the top which is kept in place with a bracket that sits on the bottom. And you do have thermal pads making contact with both faces of the SSD. With the E18 controller though, temperature throttling isn't that big a concern anymore.

PERFORMANCE

We'd like to mention right away that this is not an apples-to-apples comparison. MSI only had the 2TB review sample to share with us and most of our SSD tests are done on 1 TB SKUs. So we'll refrain from scoring the SSD in the same league as the others. We start off with CrystalDiskMark for the synthetic benchmarks. We saw read speeds of about 7200 MBps at the very maximum on the 32 Queue-1 Thread sequential test and for write speeds, we saw 6750 MBps. Moving on to ATTO, we saw write speeds of up to 6.24 GBps and read speeds of 6.94 GBps. Switching to higher file sizes resulted in consistent performance across the board which is impressive.

To test out the dynamic pSLC cache, we decided to fill up the SSD as much as we could with random data at 10 per cent capacity intervals to see when and where the speeds would start degrading. The read speeds remained untouched till the capacity of the drive was up to 40 per cent full. Upon hitting 50 per cent, the read speeds dropped to 6.4 GBps whereas write speeds remained consistent. This went on till we had 80 per cent of the NAND capacity used up. Upon hitting 90 per cent, we saw a massive degradation in write speeds as it went down to 1.78 GBps.

So the MSI Spatium M480 will perform great when fresh out of the box, like every other SSD on the planet as long as you don't cross 90% of the NAND capacity.

VERDICT

The Spatium M480 marks a stellar entry for MSI into the storage segment with transfer speeds that cross the 7000 MBps mark providing serious competition to the likes of Samsung, WD and others. Like we mentioned earlier, we can't compare the 2TB Gen 4 unit with the other 1TB Gen 4 units we've compared but if you're simply looking at the numbers, you've got a pretty great performance in the Spatium M480.

–Mithun Mohandas

SPECIFICATIONS
SPATIUM-M480-2TB | FORM FACTOR: M.2 2280 | COST/GB: Rs. 20.17 | DIMENSIONS: 80.4 x 23 x 20.4 mm | CONTROLLER: Phison E18 | Micron 96-layer TLC 3D-NAND | WARRANTY: 5 years

CONTACT
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